Adarsh Tamiri

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.10
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2020	83.70%
Sri Chaitanya Techno School	Class X (Telangana State Board of Secondary Education)	2018	9.30%

PROFILE SUMMARY

Aerospace Engineering graduate specializing in structural analysis, aerospace design, and simulation-driven engineering. Skilled in structural sizing, numerical modeling, and multidisciplinary analysis, with a focus on evaluating structural performance and supporting design optimization for aerospace applications.

TECHNICAL SKILLS

Structural Analysis: Static Stress, Linear Buckling, Nonlinear Static Analysis, Design Optimization, Flutter / Dynamic Aeroelasticity, Sturctural Sizing | MSC Patran, Ansys Mechanical, Abaqus

CAD & Modeling: Parametric Modeling, Surface Modeling, GD&T | SolidWorks, Fusion, Salome, CATIA

Programming: C++, Python, MATLAB, Bash Scripting

WORK EXPERIENCE

Industrial Trainee

Aircraft Research & Design Centre (ARDC), Hindustan Aeronautics Limited (HAL), Bangalore, India

- Evaluated structural assembly sequences for aircraft pylon and airframe integration, optimizing assembly workflows and manufacturing tolerances.
- Reviewed airworthiness certification documentation and composite material structural testing protocols alongside senior airframe design teams.
- Modeled parametric primary wing pylon components in CATIA V5, strictly complying with **AS9100 Revision D** quality and GD&T specifications.

TECHNICAL PROJECTS

Predicting Remaining Useful Life (RUL) in Turbofan Engines

Hyderabad, Telangana, India

- Engineered an end-to-end predictive maintenance pipeline on 21-sensor NASA C-MAPSS telemetry data to estimate jet engine Remaining Useful Life (RUL).
- Processed raw sensor signals using Exponential Moving Average (alpha = 0.3) filtering and 15-cycle rolling statistics to eliminate multi-regime noise.
- Trained an Asymmetric Risk Surrogate model, lowering NASA's official **Penalty Score by 43.4%** (S = 581.94) with a precision of ~19 RMSE cycles

Finite Element Structural Sizing of Transport Aircraft Fuselage Barrel

Hyderabad, Telangana, India

- Constructed a 4 m semi-monocoque fuselage FE model in MSC Patran using CQUAD4 shell and CBAR beam elements under internal pressure, shear, and bending loads.
- Performed static stress (SOL 101) and linear eigenvalue buckling (SOL 105) runs in MSC Nastran for Al 2024-T3, verifying structural safety with yield margin MS = +0.243 and buckling margin MS = +0.400.
- Conducted parametric skin thickness sweeps (1.2 mm to 2.0 mm), identifying 1.6 mm as the optimal structural baseline to eliminate yield failure while **minimizing airframe weight**.

CERTIFICATIONS

Digitalization in Aerospace Industry

Scientific Computing with Python

Machine Learning with Python

RELEVANT COURSEWORK

Indian Institute Of Technology Hyderabad, Kandi, Telangana, India

AE5120: Flight Dynamics and Structures

AE5101: Aerospace Computational Laboratory

ME5010: Mathematical Methods for Engineers

AE5111: Graduate Laboratory for Aerospace Engineering

ME7100: Advanced Topics in Mathematical Tools

LA5180: Advanced Communication Skills